

and Vishay Components India) to revamp their existing facilities to cater to this demand for chip-based diodes. According to industry experts, this forward thinking by the Indian manufacturers would help drive the Indian diode market in the future.

There has also been an increase in the demand for non-leaded diodes compared to the last few years. With the applications industries moving to surface-mount technology (miniaturisation) for production of PCBs, the demand for non-leaded diodes is expected to grow in the next few years.

Types of diodes

Diodes are classified on the basis of the configuration of the p-n junction. Some of the popular classifications of diodes are listed in Fig. 2.

Diodes are generally classified as low-power, medium-power and high-power diodes, in standard recovery characteristics and fast recovery characteristics. Besides that, there are several different diodes that are used as rectifiers, such as zener diodes, RF diodes, LEDs, Schottky rectifiers and various surface-mount type diodes.

The market value of rectifier diodes is estimated to have reached \$96 million in FY2011-12 from \$70 million in FY2009-10. The market for zener diodes is estimated to have reached \$29 million, while the market for other diodes has increased marginally to \$15 million.

Vishay Semiconductors' diodes portfolio includes Schottky, switching, PIN and Zener diodes, as well as products for transient voltage suppression, electrostatic discharge protection, and electromagnetic interference (EMI) filtering. Rectifiers from Vishay include patented TMBS and FRED Pt devices in low-profile eSMP packages, which are said to save space, reduce power losses, and improve efficiency in computing, automotive, industrial, telecommunications and other applications.

CDIL manufactures and supplies diodes in leaded and surface-mount packages. Its diodes portfolio consists of switching diodes, general-purpose

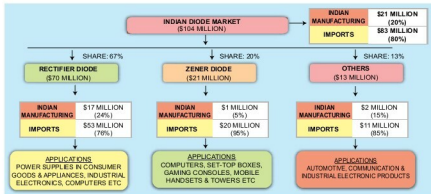


Fig. 2: Types of diodes (Source: ELCINA's ELCOMOS report)

TABLE IV
An Overview of Indian Transistors Market

Heads	Description
Overall Indian market size in FY 2009-10	\$83 million
Ratio of Imports : Indian manufacturing in FY 2009-10	80 : 20
Growth in FY 2009-10	1 per cent
Export in FY 2009-10	\$3 million
Market size by type of transistors	Bipolar junction transistors (BJTs), field-effect transistors (FETs), metal-oxide field-effect transistors (MOSFETs), insulated gate bipolar transistors (IGBTs)
Key application segments	Consumer durable and lighting
Estimated growth in FY 2010-11	12 per cent
Estimated ratio of imports : Indian manufacturing in FY 2010-11	70 : 30
Source: ELCINA's ELCOMOS report	

diodes and Schottky diodes.

Naina Semiconductor's diode portfolio includes axial leaded diodes, power diodes, auto diodes, zener diodes and Schottky diodes. It has launched a super-fast recovery dual diode module featuring high surge current capability, low forward voltage drop and super fast switching. It is targeted at such applications as power supplies, battery chargers, induction heating, solar systems and automotive systems.

Ruttonsha International Rectifier's diodes portfolio includes standard recovery and fast recovery diodes.

Transistors market

According to ELCINA's ELCOMOS report, the Indian transistor market

was estimated at \$83 million in FY2009-10 with imports accounting for 80 per cent of the total market. Transistors also are largely imported from China and Taiwan. In addition, a considerable proportion of transistors used in automotive and power electronics is imported from Europe owing to their high quality.

It is estimated that there are more than 20 transistor manufacturing units operating in India,

with 80 per cent of them being classified under small-scale industry. The industry is largely dominated by traders who import transistors from China and Taiwan. The two dominant players in the Indian market are Continental Devices India Limited (CDIL) and Bharat Electronics Limited (BEL). In addition, other companies like Semikron and Arihant manufacture IGBTs for niche industrial products.

CDIL is a manufacturer and supplier of transistors in leaded and surface-mount packages. Its transistors portfolio consists of GP switching transistors, small signal transistors, medium-power transistors, power transistors, high-voltage transistors, low-frequency transistors, RF transistors and Darlington transistors.